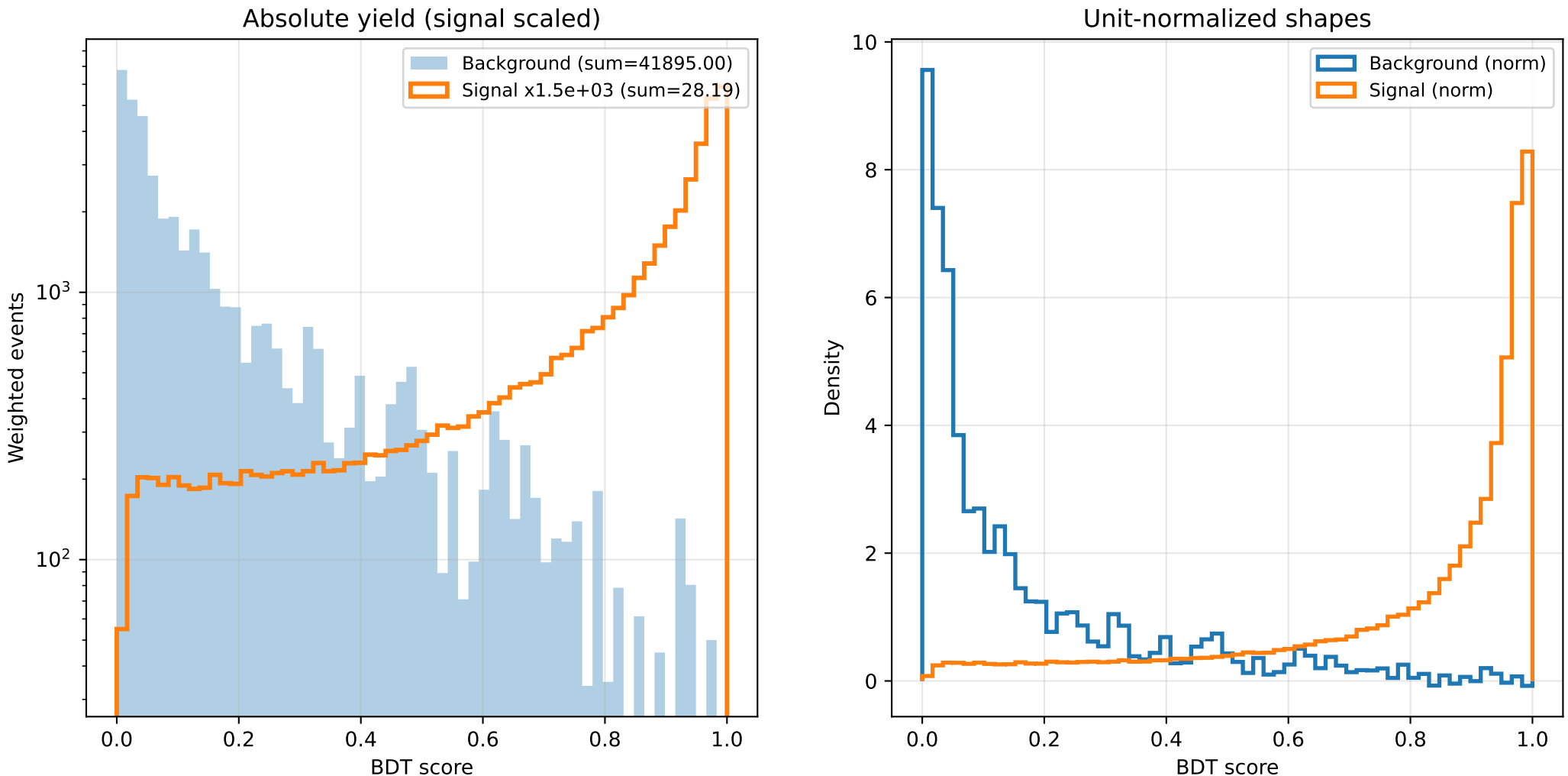
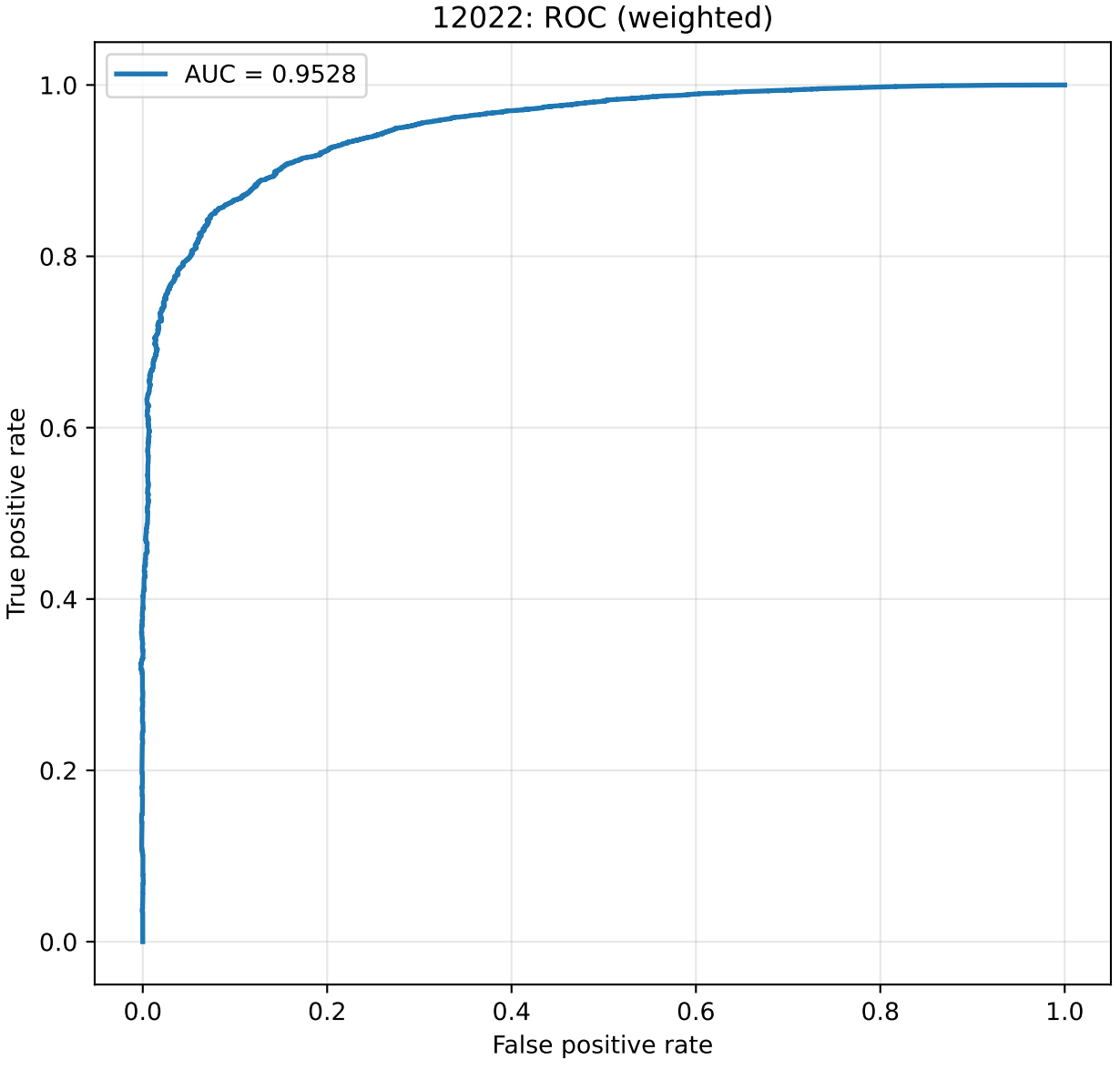
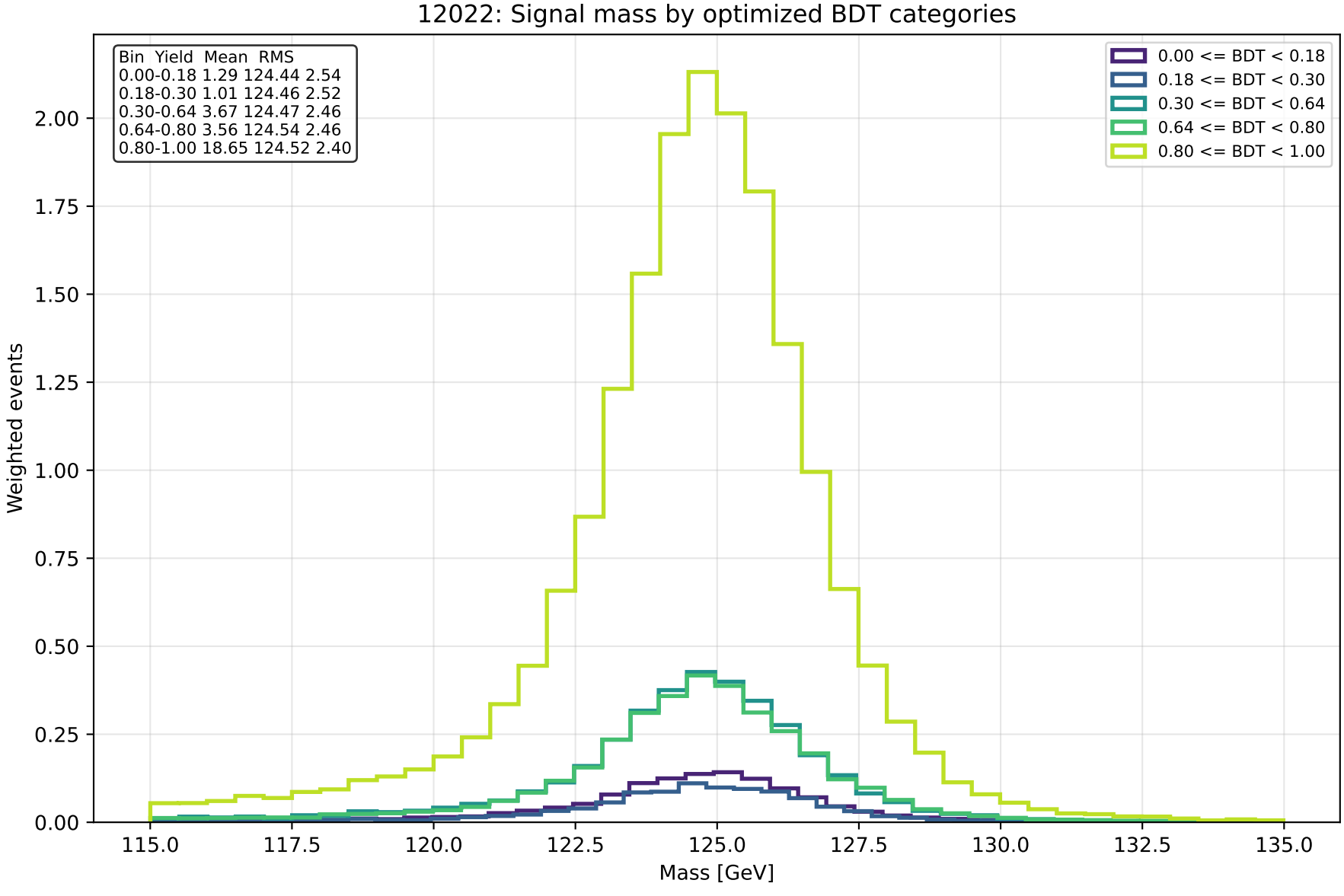
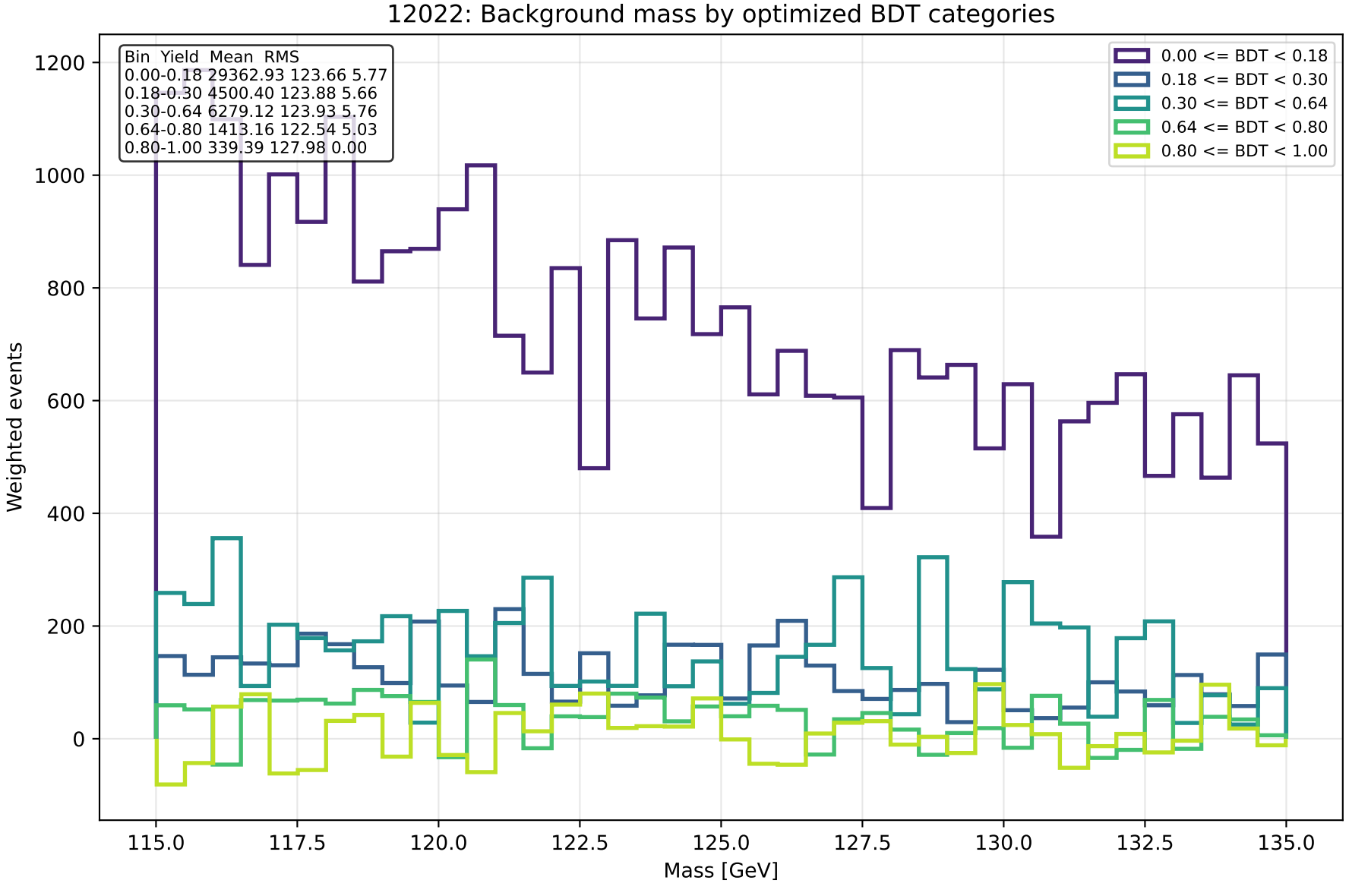

12022: BDT scores in 115-135 GeV
Absolute yield (signal scaled)
Background (sum=41895.00)
Signal x1.5e+03 (sum=28.19)
Weighted events
10^3
10^2
0.0
0.2
0.4
0.6
0.8
1.0
BDT score
Unit-normalized shapes
Background (norm)
Signal (norm)
Density
10
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
BDT score

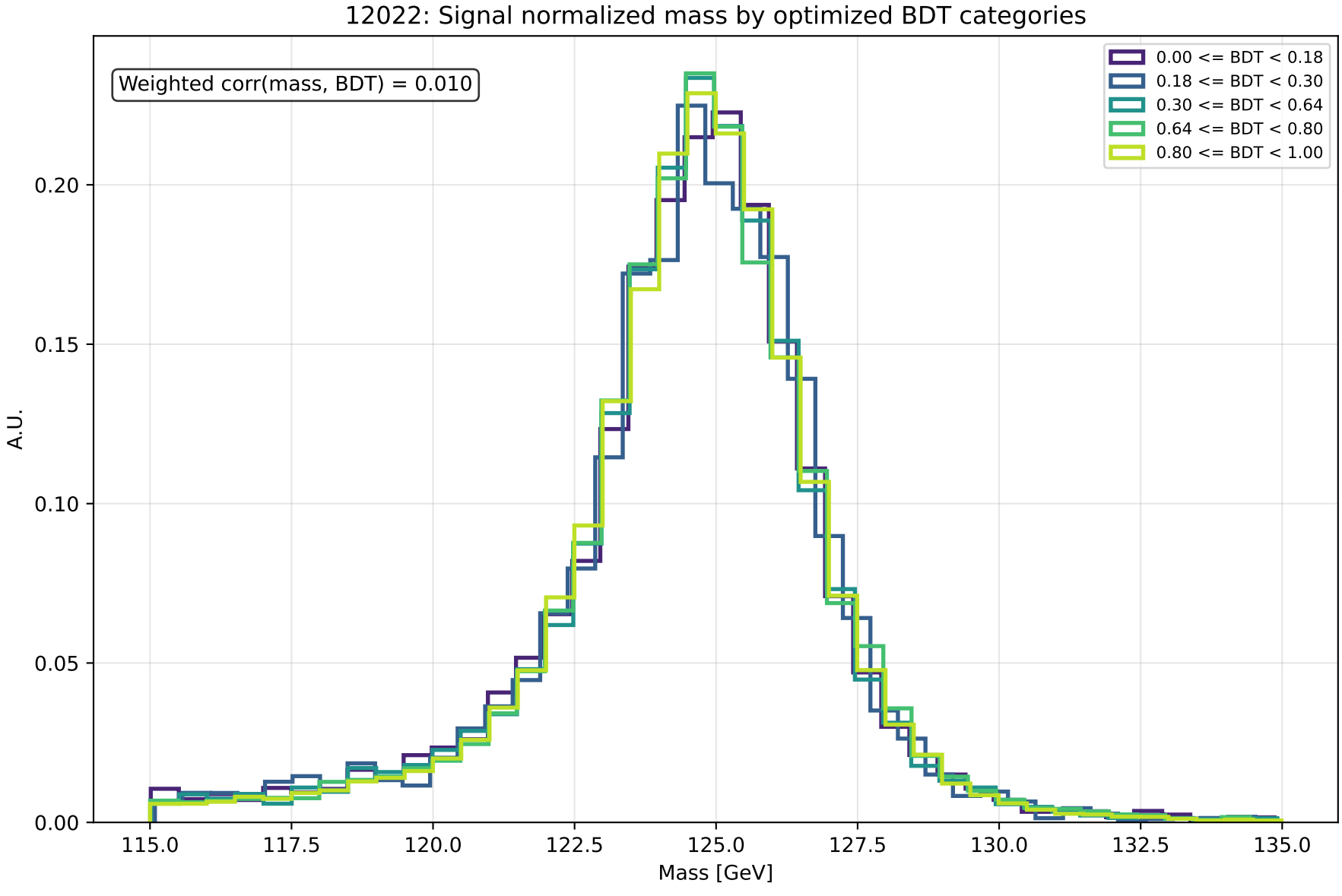

12022: ROC (weighted)
AUC = 0.9528
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0

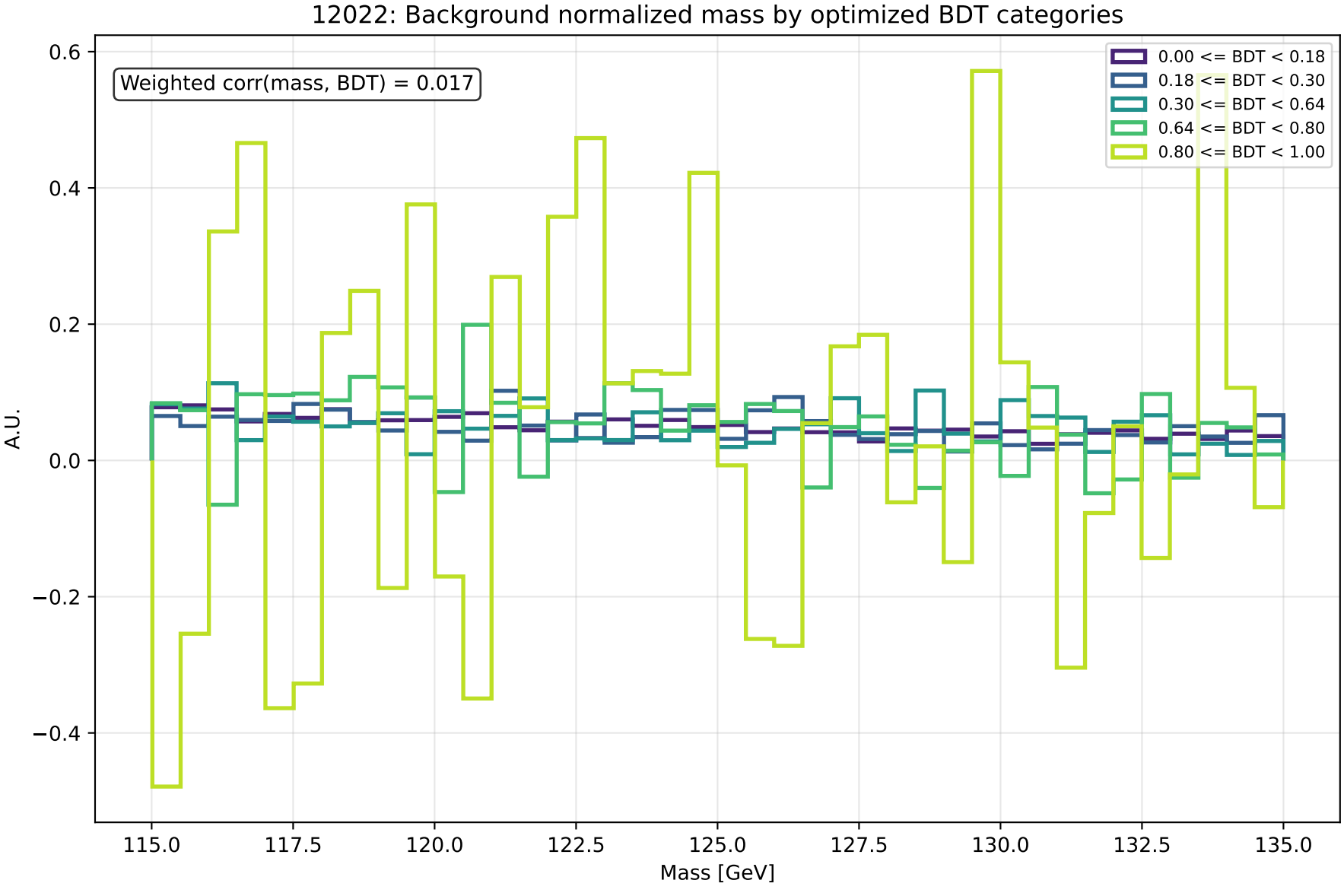
12022: Multi-bin BDT categorization (no hard cut)
Optimized N=5, Zexp(profile)=23.506, monotonic S/B=True

Bin	BDT range	S	B	S/B	S/sqrt(B)
1	0.000-0.180	1.286	29362.930	4.38e-05	0.008
2	0.180-0.304	1.013	4500.401	0.000225	0.015
3	0.304-0.637	3.674	6279.119	0.000585	0.046
4	0.637-0.798	3.563	1413.163	0.00252	0.095
5	0.798-1.000	18.649	339.385	0.0549	1.012


12022: Signal mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.18 1.29 124.44 2.54
0.18-0.30 1.01 124.46 2.52
0.30-0.64 3.67 124.47 2.46
0.64-0.80 3.56 124.54 2.46
0.80-1.00 18.65 124.52 2.40
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]


12022: Background mass by optimized BDT categories
Bin Yield Mean RMS
0.00-0.18 29362.93 123.66 5.77
0.18-0.30 4500.40 123.88 5.66
0.30-0.64 6279.12 123.93 5.76
0.64-0.80 1413.16 122.54 5.03
0.80-1.00 339.39 127.98 0.00
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
200
400
600
800
1000
1200
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0


12022: Signal normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

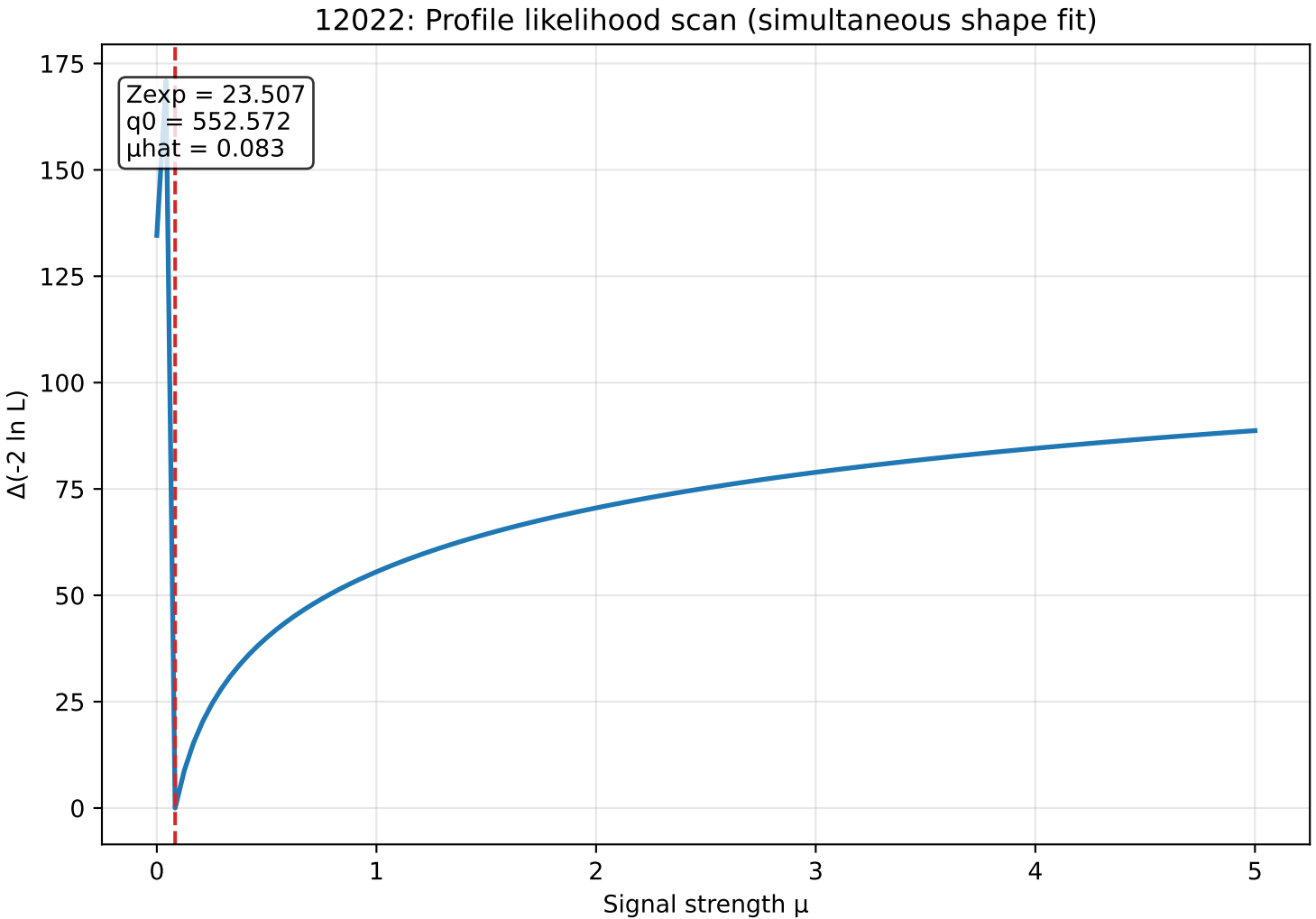

12022: Background normalized mass by optimized BDT categories
Weighted corr(mass, BDT) = 0.017
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.6
0.4
0.2
0.0
−0.2
−0.4
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

12022: Simultaneous shape-based profile likelihood summary

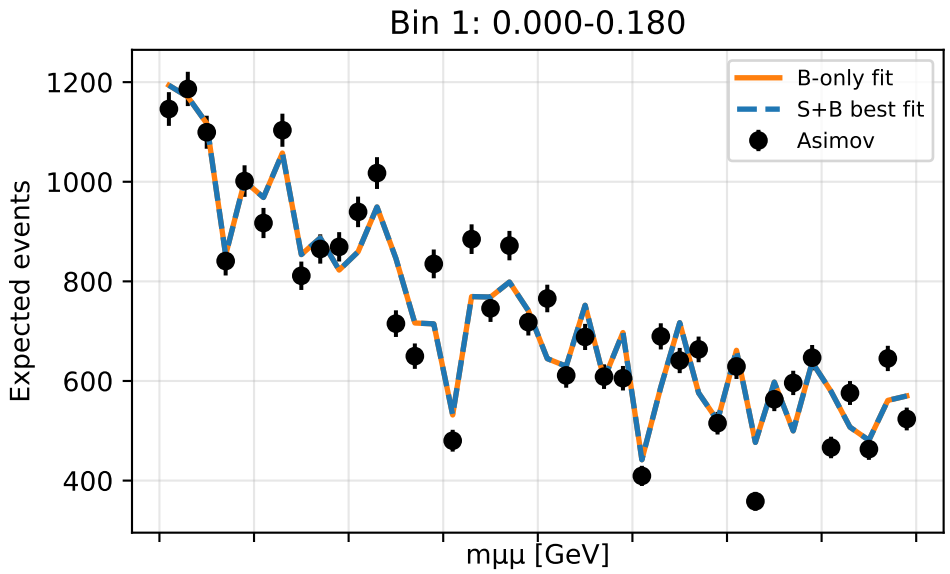
Quantity	Value
Expected Z	23.5069
q0	552.5722
mu_hat	0.0833
N_BDT bins	5
N_mass bins	40
poly order	2
bkg norm unc	0.200
poly unc	0.300
core poly order	2
core from low/med bins	True
DY >=2j unc	0.050
sig norm unc	0.050
sig mass shift unc [GeV]	0.200
sig resolution unc	0.100

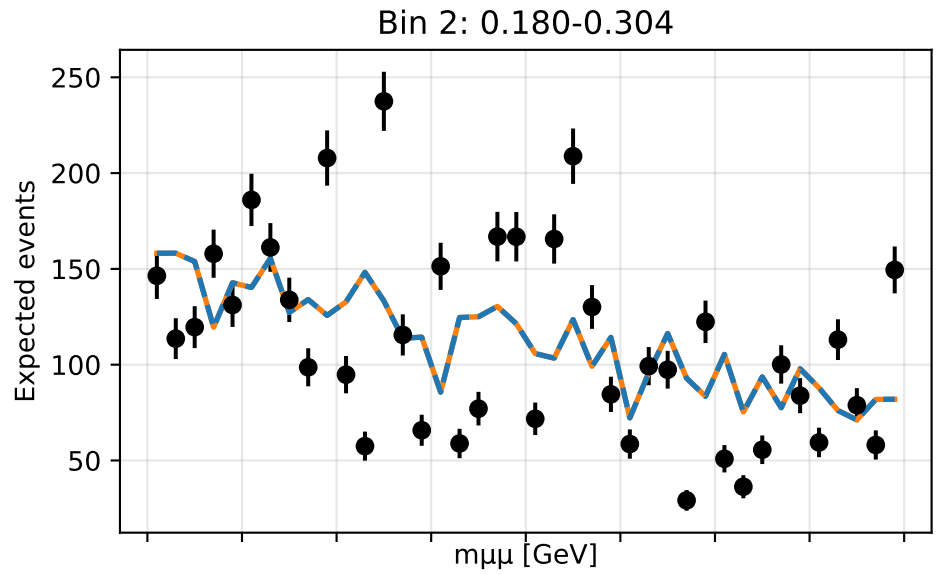
12022: Shape-likelihood bin yields (Asimov)

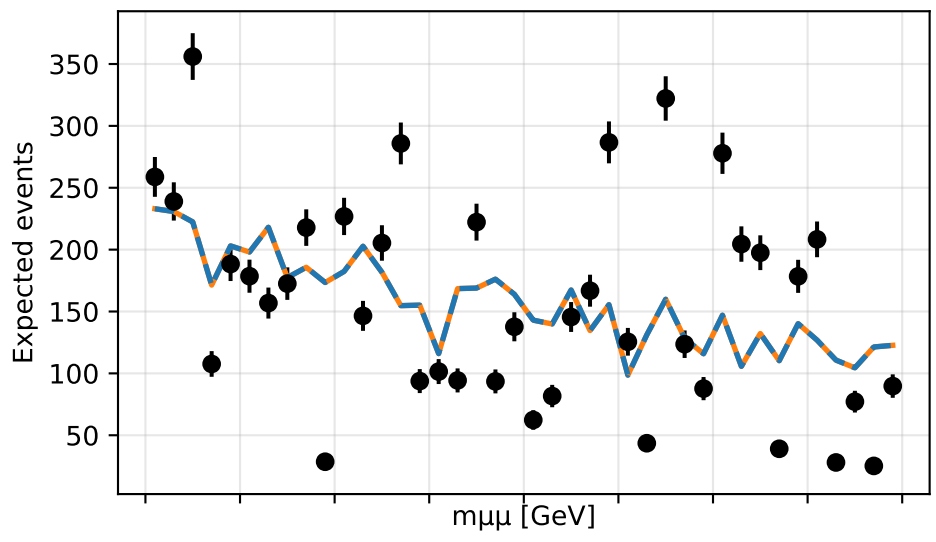
Bin	BDT	S truth	B truth	B-only fit	S+B fit
1	0.000-0.180	1.286	29362.930	29362.930	29363.010
2	0.180-0.304	1.013	4500.401	4500.401	4500.480
3	0.304-0.637	3.674	6279.119	6279.119	6279.199
4	0.637-0.798	3.563	1413.163	1413.163	1413.242
5	0.798-1.000	18.649	339.385	358.314	369.305

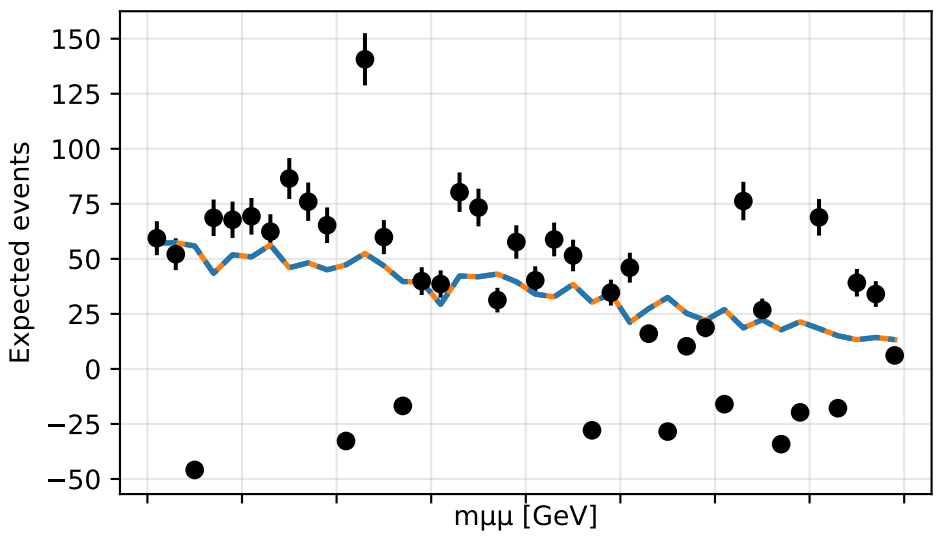

12022: Profile likelihood scan (simultaneous shape fit)
Zexp = 23.507
q0 = 552.572
μhat = 0.083
Δ(-2 ln L)
Signal strength μ
175
150
125
100
75
50
25
0
0
1
2
3
4
5

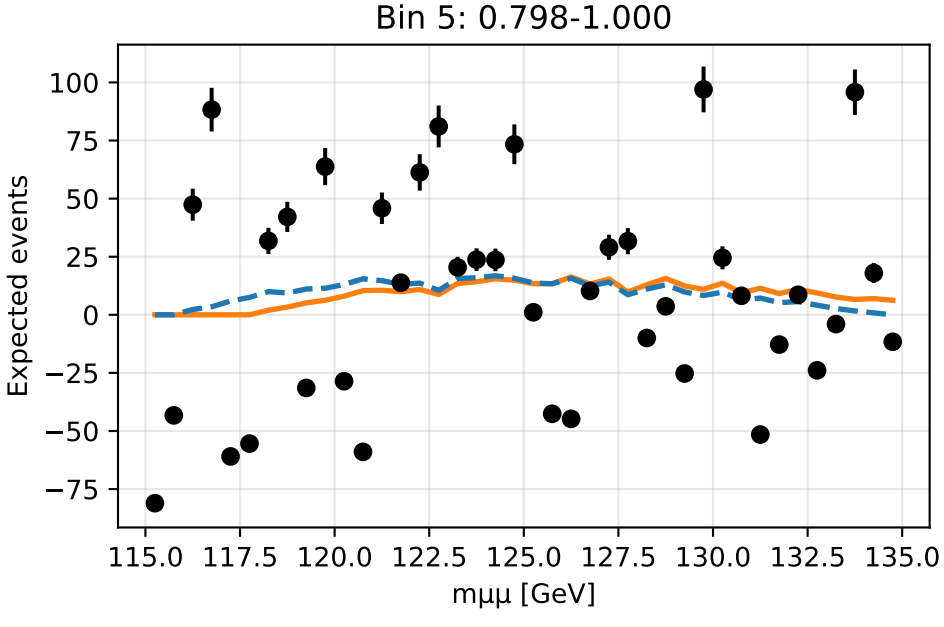
12022: Asimov mass templates by BDT bin (simultaneous fit)









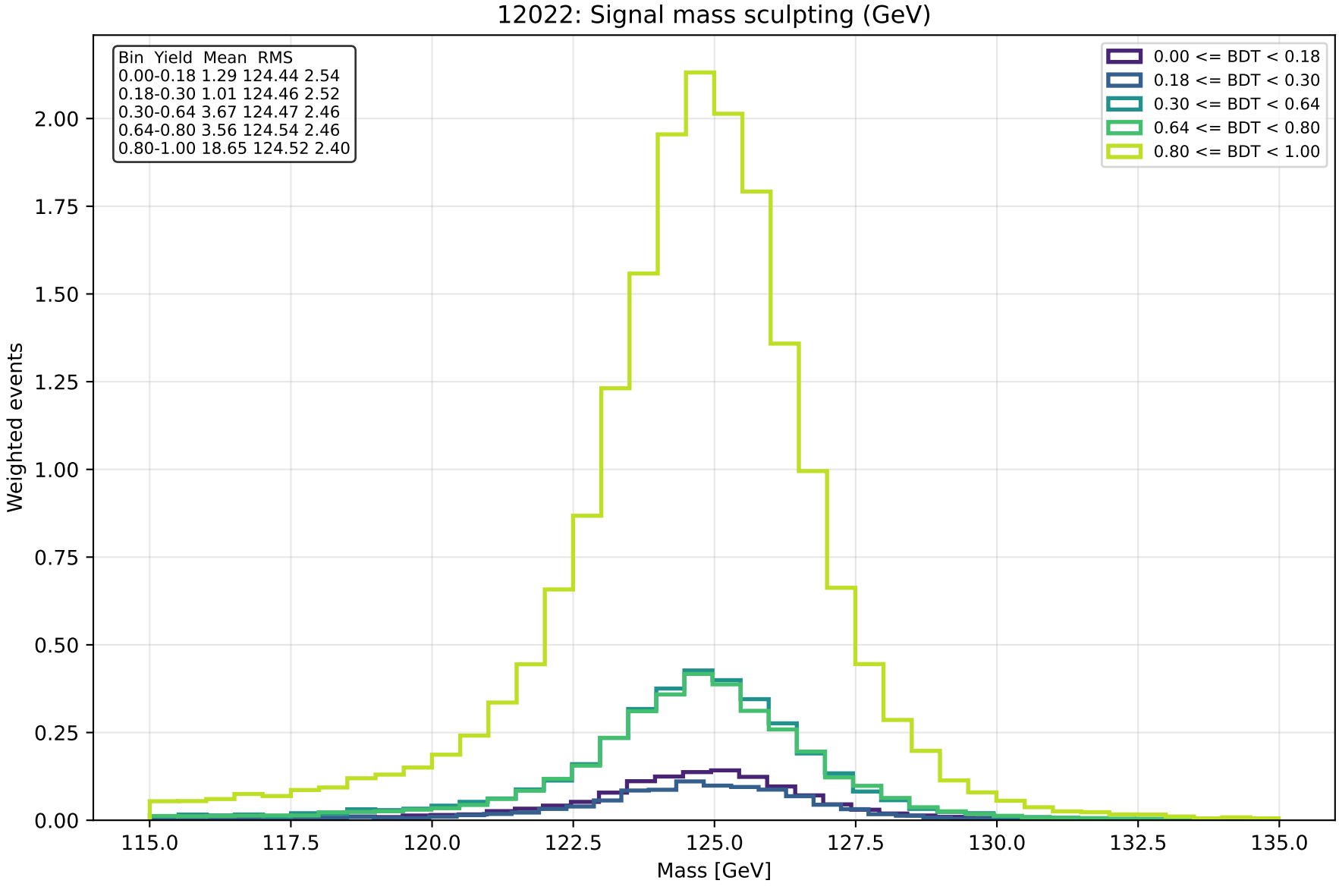


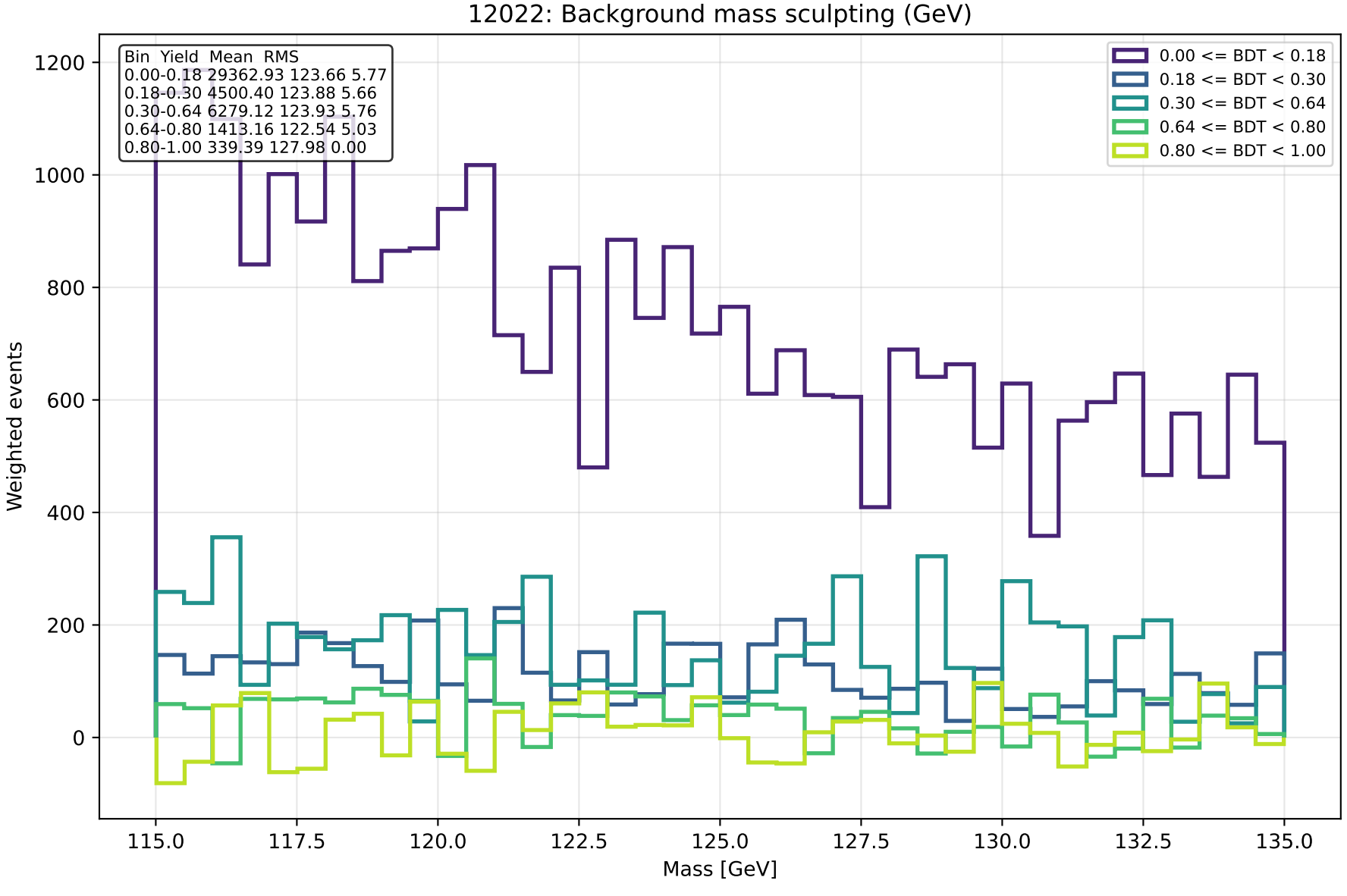
SumExp	LandaUxBernstein	0.0000	0.0000	-1.0000	FAIL
SumExp	BWZxBernstein	3.0000	4.5124	2.0000	FAIL
SumExp	BWZGamma	7.8000	7.3999	6.8000	FAIL
PowerLaw	BWZRedux	2.0083	2.5761	1.0083	FAIL
PowerLaw	SumExp	10.0050	4.0403	9.0050	FAIL
PowerLaw	PowerLaw	3.3167	3.4389	2.3167	FAIL
PowerLaw	FEWZxBernstein	2.1850	2.6615	1.1850	FAIL
PowerLaw	LandaUxBernstein	0.2833	1.1608	-0.7167	FAIL
PowerLaw	BWZxBernstein	2.6667	3.1555	1.6667	FAIL
PowerLaw	BWZGamma	8.3967	5.5984	7.3967	FAIL
FEWZxBernstein	BWZRedux	2.5083	3.3769	1.5083	FAIL
FEWZxBernstein	SumExp	10.8250	5.1021	9.8250	FAIL
FEWZxBernstein	PowerLaw	3.0283	3.3740	2.0283	FAIL
FEWZxBernstein	FEWZxBernstein	2.2833	2.4033	1.2833	FAIL
FEWZxBernstein	LandaUxBernstein	0.0000	0.0000	-1.0000	FAIL
FEWZxBernstein	BWZxBernstein	2.9050	3.7505	1.9050	FAIL
FEWZxBernstein	BWZGamma	9.0517	4.4167	8.0517	FAIL
LandaUxBernstein	BWZRedux	445.9450	13.0445	444.9450	FAIL
LandaUxBernstein	SumExp	885.4650	5.7962	884.4650	FAIL
LandaUxBernstein	PowerLaw	552.9017	12.1973	551.9017	FAIL
LandaUxBernstein	FEWZxBernstein	718.0167	8.1567	717.0167	FAIL
LandaUxBernstein	LandaUxBernstein	1.1617	1.1806	0.1617	PASS
LandaUxBernstein	BWZxBernstein	795.3017	9.8894	794.3017	FAIL
LandaUxBernstein	BWZGamma	888.4967	7.1844	887.4967	FAIL
BWZxBernstein	BWZRedux	4.3083	5.4251	3.3083	FAIL
BWZxBernstein	SumExp	5.2783	4.2930	4.2783	FAIL

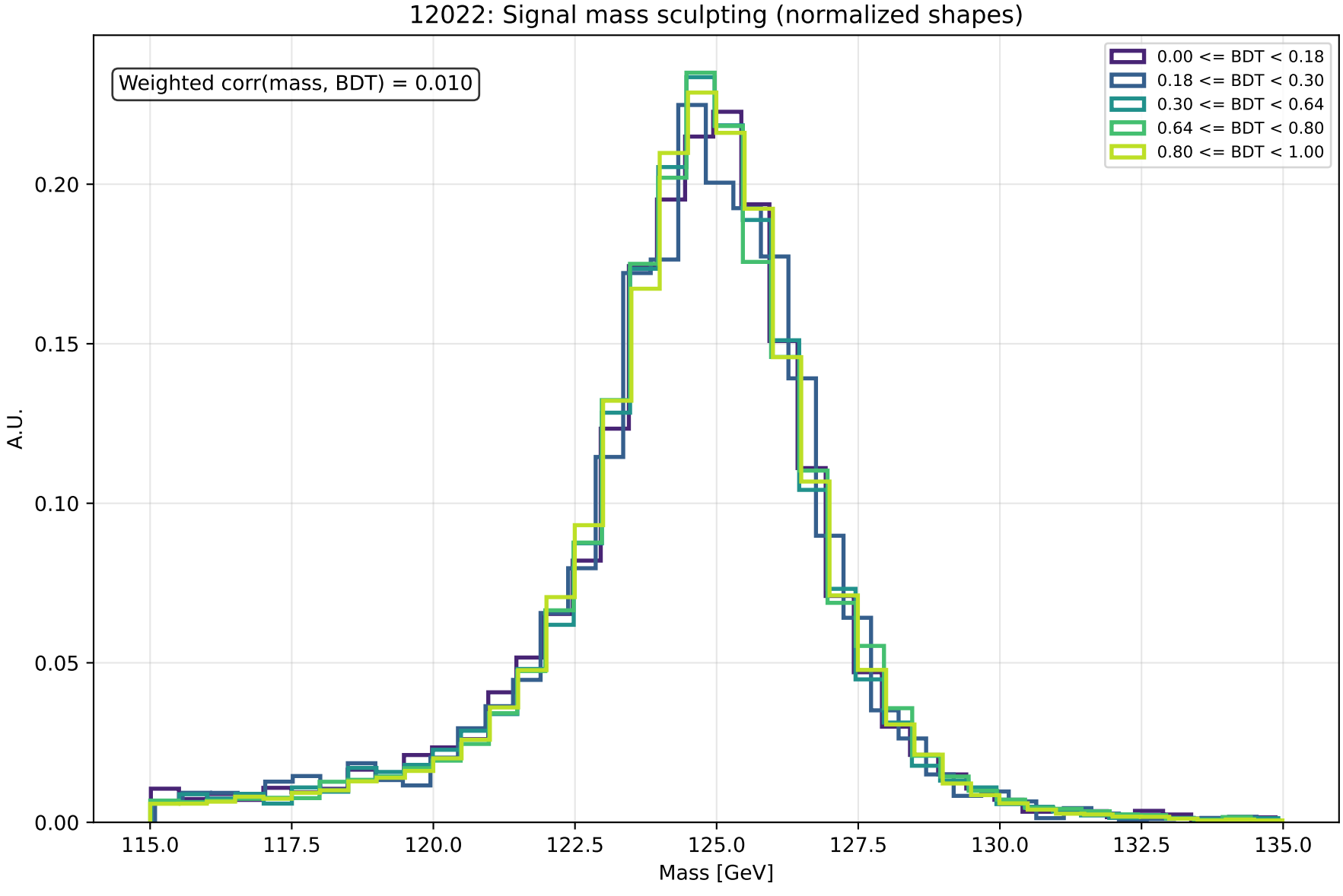
12022 Bias matrix (mu_true=1.00, threshold=0.20)

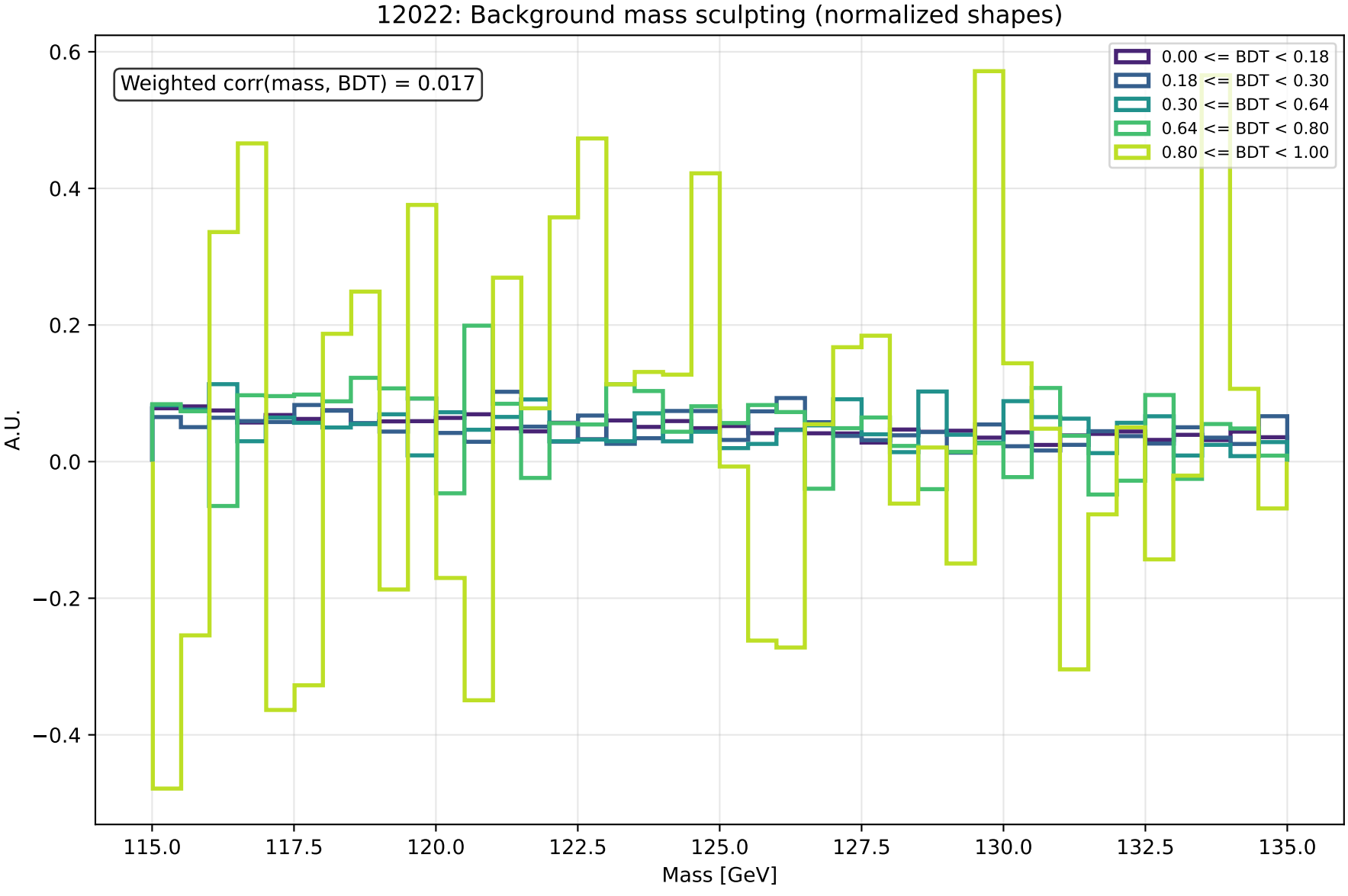
12022: Expected yield per optimized BDT bin (115-135 GeV)
S total=28.19, B total=41895.00

BDT bin	S (bin)	S frac	S>=lo	B (bin)	S/B (>=lo)
0.000-0.180	1.286	4.6%	100.0%	29362.930	0.000673
0.180-0.304	1.013	3.6%	95.4%	4500.401	0.00215
0.304-0.637	3.674	13.0%	91.8%	6279.119	0.00322
0.637-0.798	3.563	12.6%	78.8%	1413.163	0.0127
0.798-1.000	18.649	66.2%	66.2%	339.385	0.0549

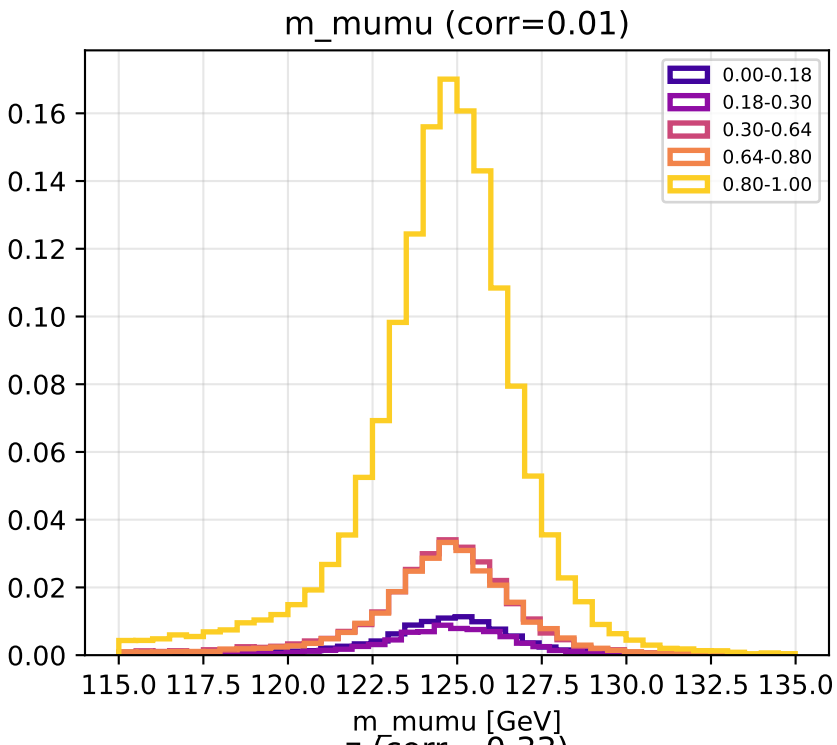

12022: Signal mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.18 1.29 124.44 2.54
0.18-0.30 1.01 124.46 2.52
0.30-0.64 3.67 124.47 2.46
0.64-0.80 3.56 124.54 2.46
0.80-1.00 18.65 124.52 2.40
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

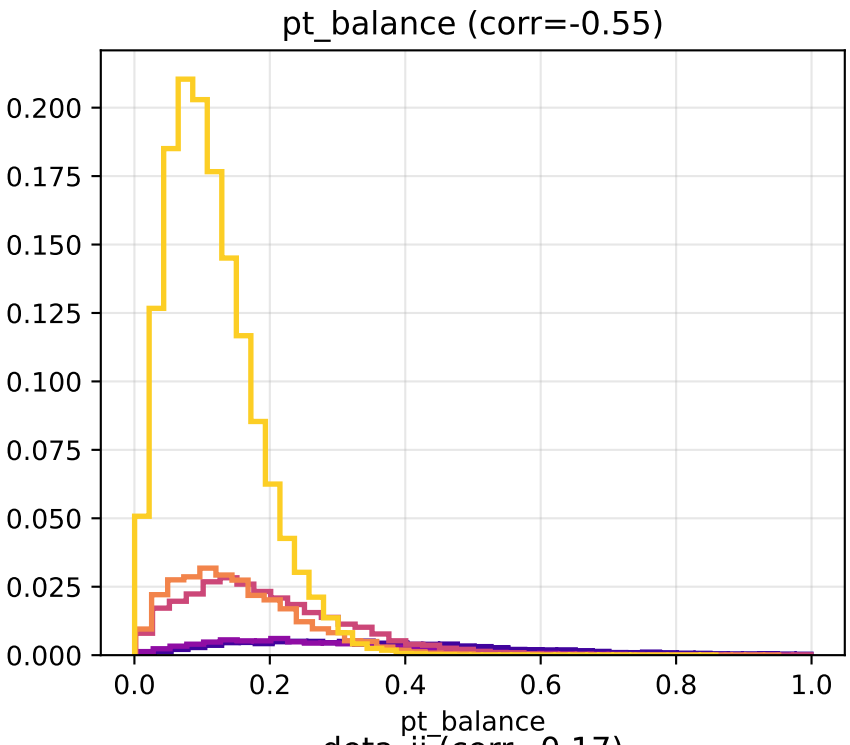

12022: Background mass sculpting (GeV)
Bin Yield Mean RMS
0.00-0.18 29362.93 123.66 5.77
0.18-0.30 4500.40 123.88 5.66
0.30-0.64 6279.12 123.93 5.76
0.64-0.80 1413.16 122.54 5.03
0.80-1.00 339.39 127.98 0.00
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
Weighted events
Mass [GeV]
0
200
400
600
800
1000
1200
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0

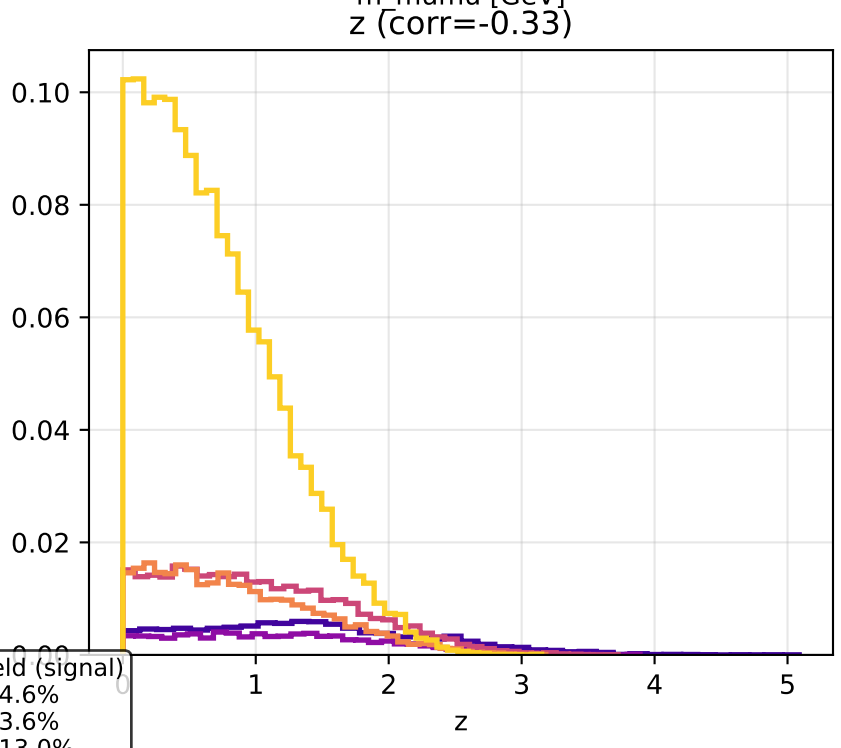

12022: Signal mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.010
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.00
0.05
0.10
0.15
0.20
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

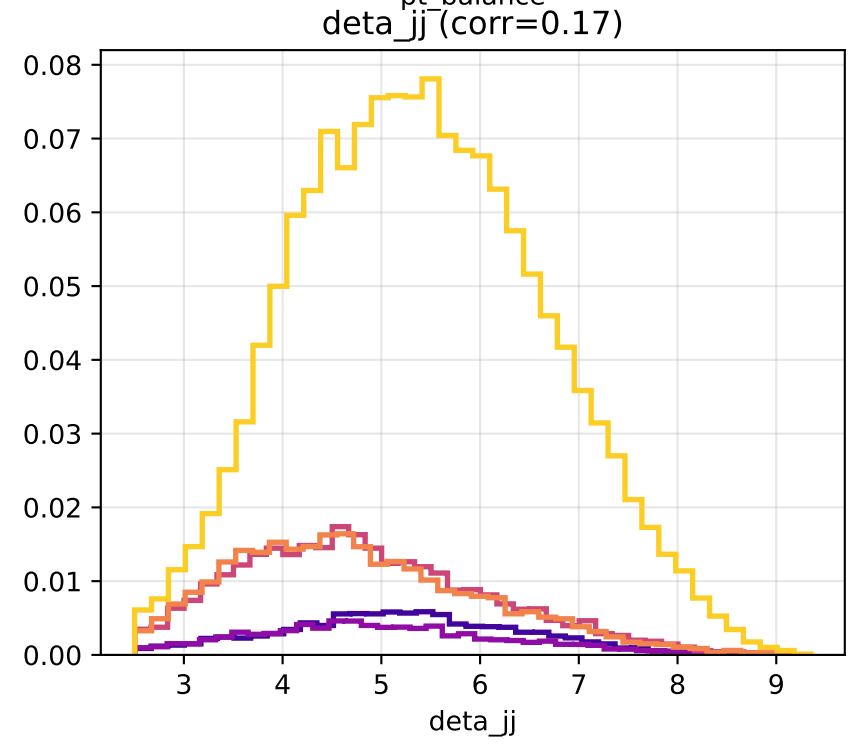

12022: Background mass sculpting (normalized shapes)
Weighted corr(mass, BDT) = 0.017
0.00 <= BDT < 0.18
0.18 <= BDT < 0.30
0.30 <= BDT < 0.64
0.64 <= BDT < 0.80
0.80 <= BDT < 1.00
A.U.
0.6
0.4
0.2
0.0
−0.2
−0.4
115.0
117.5
120.0
122.5
125.0
127.5
130.0
132.5
135.0
Mass [GeV]

12022: Signal distributions by BDT bin

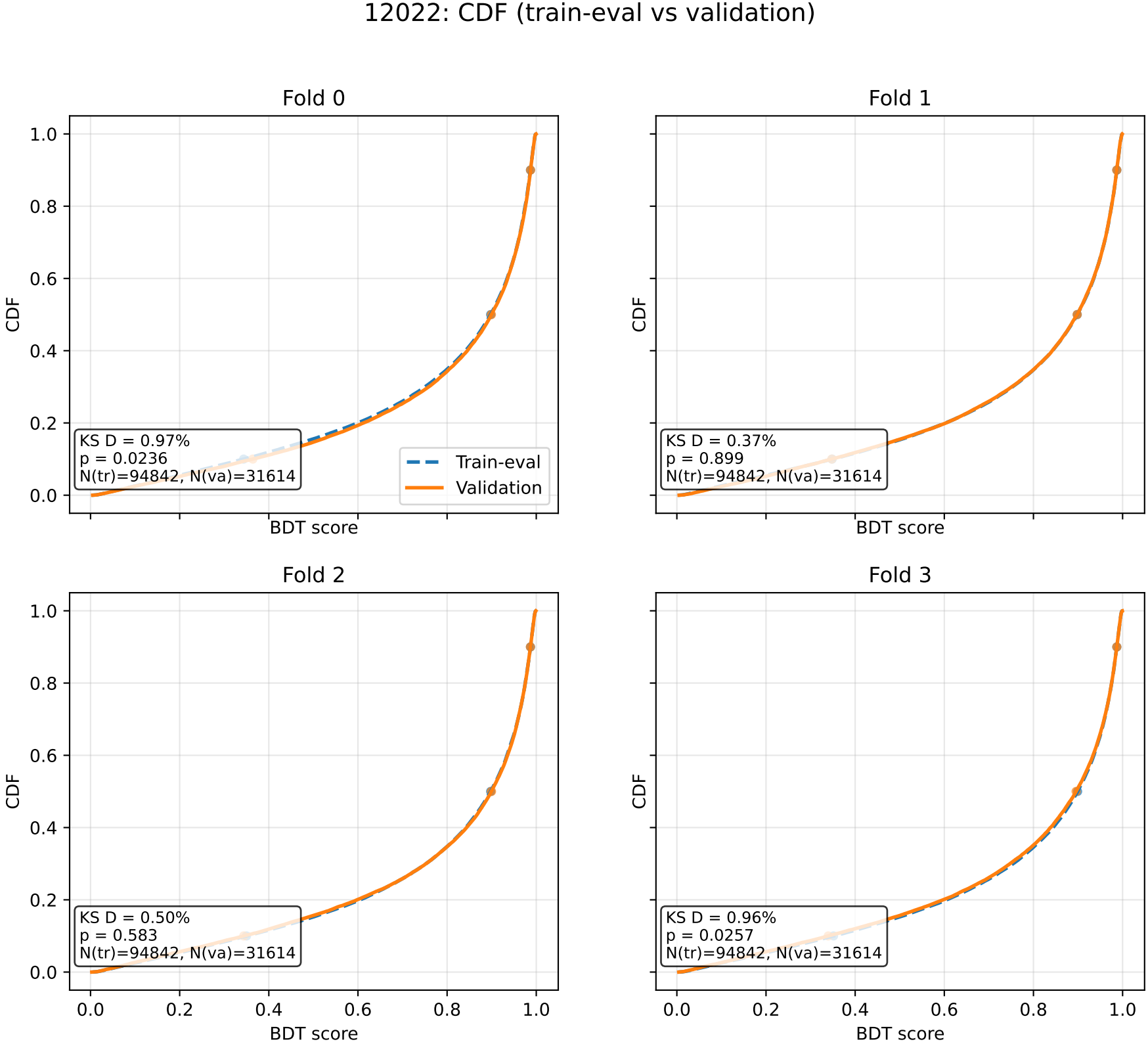








BDT bin yield (signal)
0.00-0.18: 4.6%
0.18-0.30: 3.6%
0.30-0.64: 13.0%
0.64-0.80: 12.6%
0.80-1.00: 66.2%


12022: CDF (train-eval vs validation)
Fold 0
KS D = 0.97%
p = 0.0236
N(tr)=94842, N(va)=31614
Train-eval
Validation
Fold 1
KS D = 0.37%
p = 0.899
N(tr)=94842, N(va)=31614
Fold 2
KS D = 0.50%
p = 0.583
N(tr)=94842, N(va)=31614
Fold 3
KS D = 0.96%
p = 0.0257
N(tr)=94842, N(va)=31614
CDF
BDT score
0.0
0.2
0.4
0.6
0.8
1.0

12022: KS test summary

Fold	KS statistic	N events	alpha	Pass threshold	KS pass
0	0.97%	31614	0.001	0.20%	FALSE
1	0.37%	31614	0.001	0.20%	FALSE
2	0.50%	31614	0.001	0.20%	FALSE
3	0.96%	31614	0.001	0.20%	FALSE

12022: Top-10 feature importance (gain)
12022_featureimportance_DY.txt

Feature	Gain %
pt_balance	11.97%
z	11.41%
deta_jj	10.34%
njets	9.72%
pt_j2	6.57%
dphi_mumu_jj	4.80%
jet1_eta	4.77%
costheta_CS	4.17%
pt_j1	4.06%
mu1_eta	4.01%